Quality Alloys Case Study 1

By

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September 4, 2025

“On my honor as an Aggie, I have neither given nor received unauthorized aid on this academic work.”

This is entirely my own work or—if authorized by my instructor—the work of the team I was assigned to, except as documented in a bibliography or acknowledged below.

Signature * |Nikitha Joshy|

Artificial Intelligence (AI) Acknowledgement:
Google Gemini assisted with revisions to the final part of the conclusions portion of the report, and to help create the plots and graphics in python, and with rephrasing the methods and executive summary.

Submitted to fulfill the requirement for Case Study 1
DAEN 400, Fall 2025
Department of Industrial & Systems Engineering
Texas A&M University

Executive Summary

This project used Python to analyze web analytics data from Quality Alloys (QA) to assess the effectiveness of its new web presence. The analysis focused on website traffic, user engagement, and sales data to provide a comprehensive summary of QA's performance as a company after their website launch and offer actionable recommendations.

The analysis revealed that while QA's marketing efforts successfully drove a significant amount of traffic to the website, particularly during promotion periods, this traffic was not sustainable. The website initially had a good reach, with the highest percentage of visitors surprisingly coming from South America. However, user engagement was low, with a high bounce rate during promotions, indicating that visitors were quickly leaving the site after viewing only one page. This lack of engagement meant that the increased web traffic did not lead to a lasting increase in sales (pounds of alloys sold). In fact, the pre-promotion period saw the greatest number of pounds sold, which declined significantly after the promotion.

Based on these findings, it is recommended that QA focus on improving website engagement. Enhancing the landing page, making information more accessible, and adding features like an online shopping cart could encourage visitors to spend more time on the site. Furthermore, tailoring marketing content to the site's largest demographic: South American visitors, by adding Spanish-language articles or ads could help convert visitors into customers. The analysis also showed that a significant number of visitors came from googleads.g.doubleclick.net, suggesting that replicating similar marketing strategies on other platforms could effectively drive more web traffic, leading to an increase in both sales and revenue.

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1. Introduction

1.1 Introduction to Quality Alloys

Quality Alloys, Inc. (QA) is an American distributor of alloys that are used in industrial engineering, and these alloys come in multiple different grades and are sold in small quantities to buyers in lbs. Due to the project-by-project purchases, one singe purchase is not a good way to anticipate future sales. Overall, QA is a very respected and prominent company in the alloys market.

1.2 Project Goals and Objectives

The data provided by QA was from 5/25/2008 – 08/29/2009 and included information on weekly website visits, financials, pounds sold, daily site visits, and demographics. The dataset also indicates the initial period of their web presence, pre promotion, promotion, and post promotion. Currently QA's marketing strategy includes direct mail, advertising via magazines, listings on two different industrial product web portals, and a web presence. Given this information, the task of the project was to conduct analysis on whether the website is driving sales and revenue.

2. Methods

Several different analyses were conducted to gauge how well QA's website was doing in bringing in customers and increasing sales. An analysis of weekly site visits and unique visits was done to understand the overall traffic to the website over time and identify if new people were regularly visiting. It gave a good understanding of the site's reach.

Analyzing bounce rate and new visits by period was done to see if a specific promotion period was effective. The goal was to determine if the promotion helped keep visitors from leaving the site quickly (lowering the bounce rate) and if it successfully attracted new customers.

Performance metrics by period were analyzed to see if visitors were engaged enough to spend a significant amount of time on the site. This helps gauge whether marketing efforts not only bring people to the site but also keep them there, which is a key indicator of effective content.

To understand the connection between engagement and customer action, an analysis of inquiries and time spent on site was chosen. The purpose was to determine if people spending more time on the website led to more inquiries, which could indicate a strong link between the website and potential sales.

For QA, seeing the correlation between both weekly profit and the number of pounds sold would be helpful in concluding whether just selling more alloys would increase their profit. Especially since QA themselves will have costs associated with each sale, so instead of revenue, profit compared to the amount of alloys sold could provide a good estimate of whether it is beneficial for them to focus more on pushing sales regardless of costs they may incur in the process. Computing how many lbs. were sold each period can visually help QA see how their sales are going during, before, and after promotion.

A demographics analysis was chosen to help QA understand its audience. By visualizing the countries of visitors, they can tailor their website content or marketing to better appeal to their most frequent international viewers, optimizing their online presence for their key markets.

3. Results

3.1 Weekly Site Visits and Unique Visits

Figure 1 created from the methodology used in section 2.2 can be seen below.

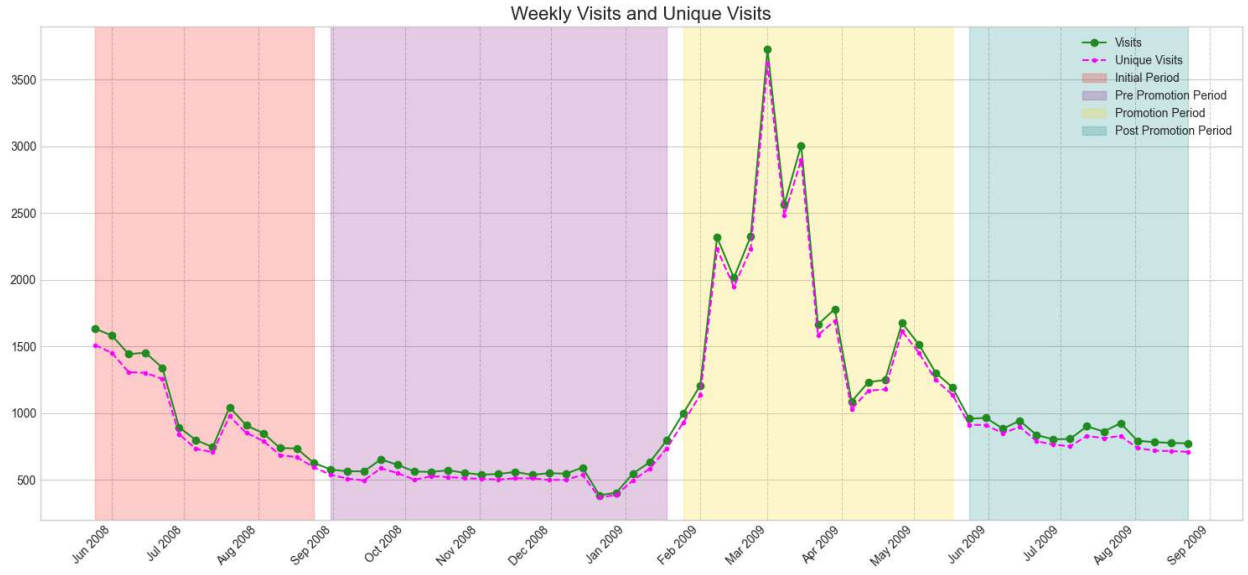


Figure 1. *Weekly Visits and Unique Visits to the QA website over 4 different periods*

3.2 Bounce Rate and New Visits by Period

Figure 2 Generated through the process explained in section 2.3 is displayed below.

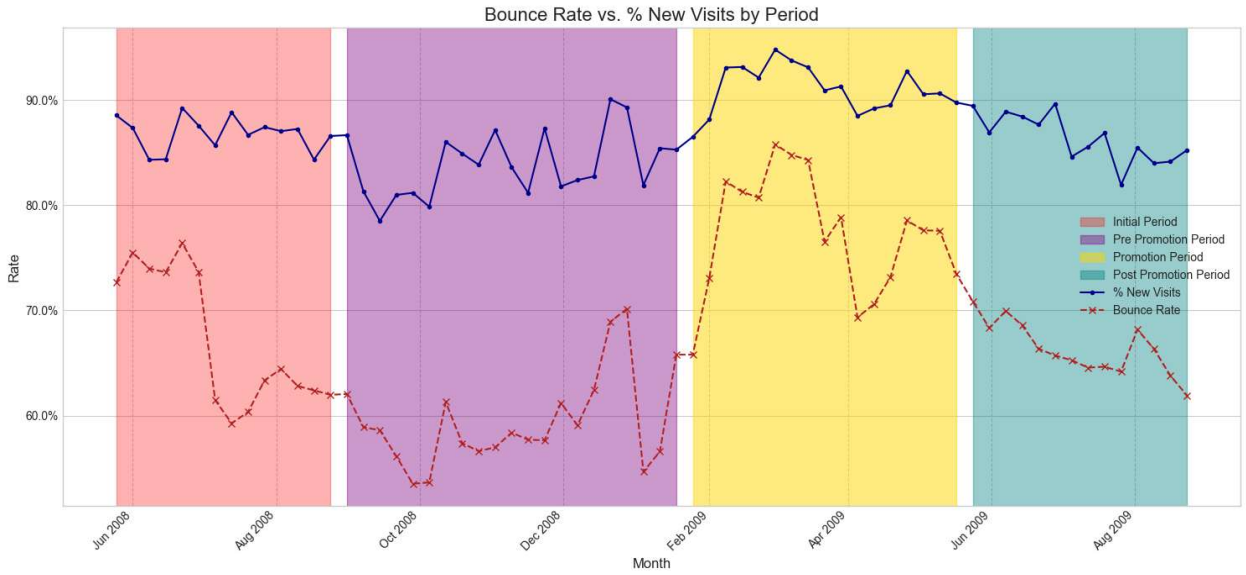


Figure 2. *Bounce rate percentage and percentage of new visits to the QA website over the course of four periods.*

3.3 Performance Metrics by Period

The performance metrics results discussed in section 2.4 can be viewed below in **Table 1**.

Table 1. *Performance Metrics by Period*

Period	Total Page Views	Avg. Pages/Visit	Time on Site (s)
Initial	32,867	2.28	80.3
Pre-Promotion	31,680	2.67	95.9
Promotion	52,731	1.79	48.8
Post-Promotion	26,155	2.18	70.0

The visual representation of **Table 1** can be seen below in **Figure 3**.

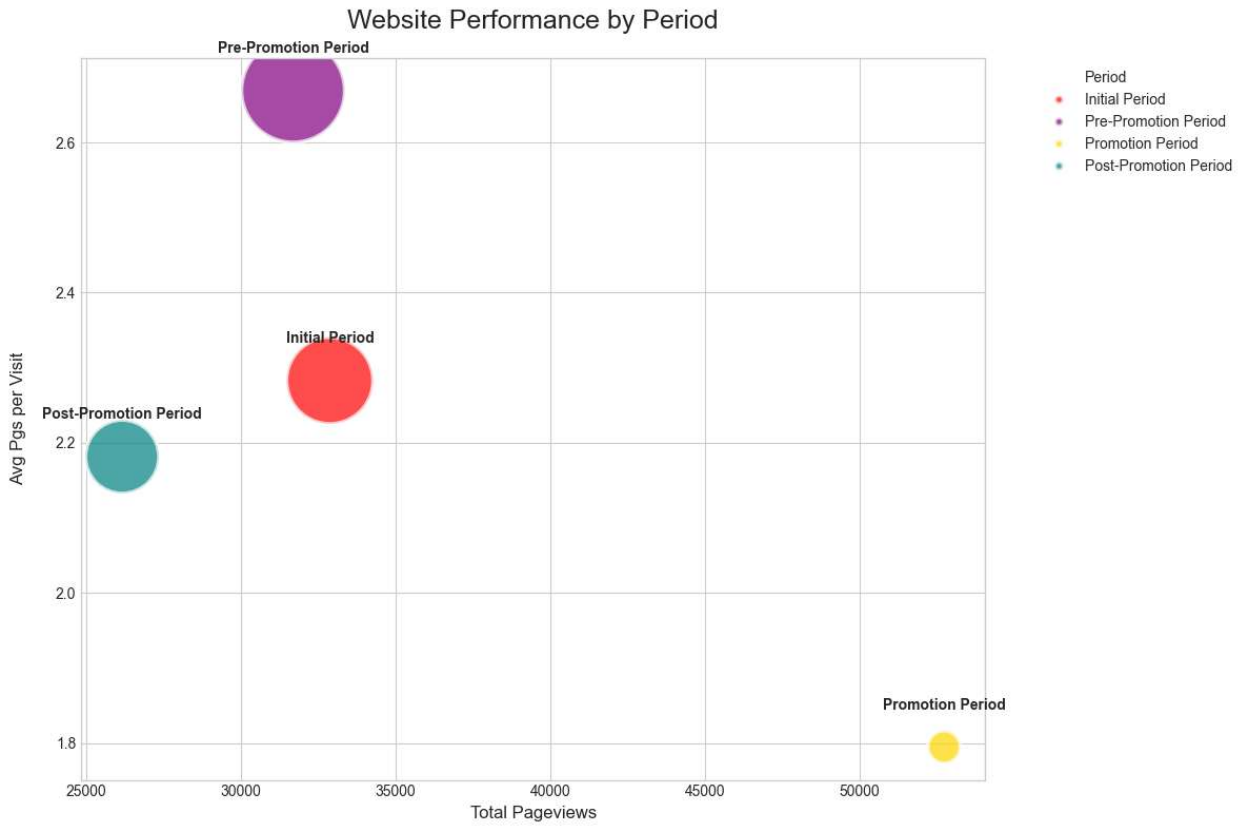


Figure 3. *Visual Representation of Performance Metrics*

3.4 Lbs. Sold

A visual representation of weekly profit and lbs. sold can be seen in **Figure 4** below.

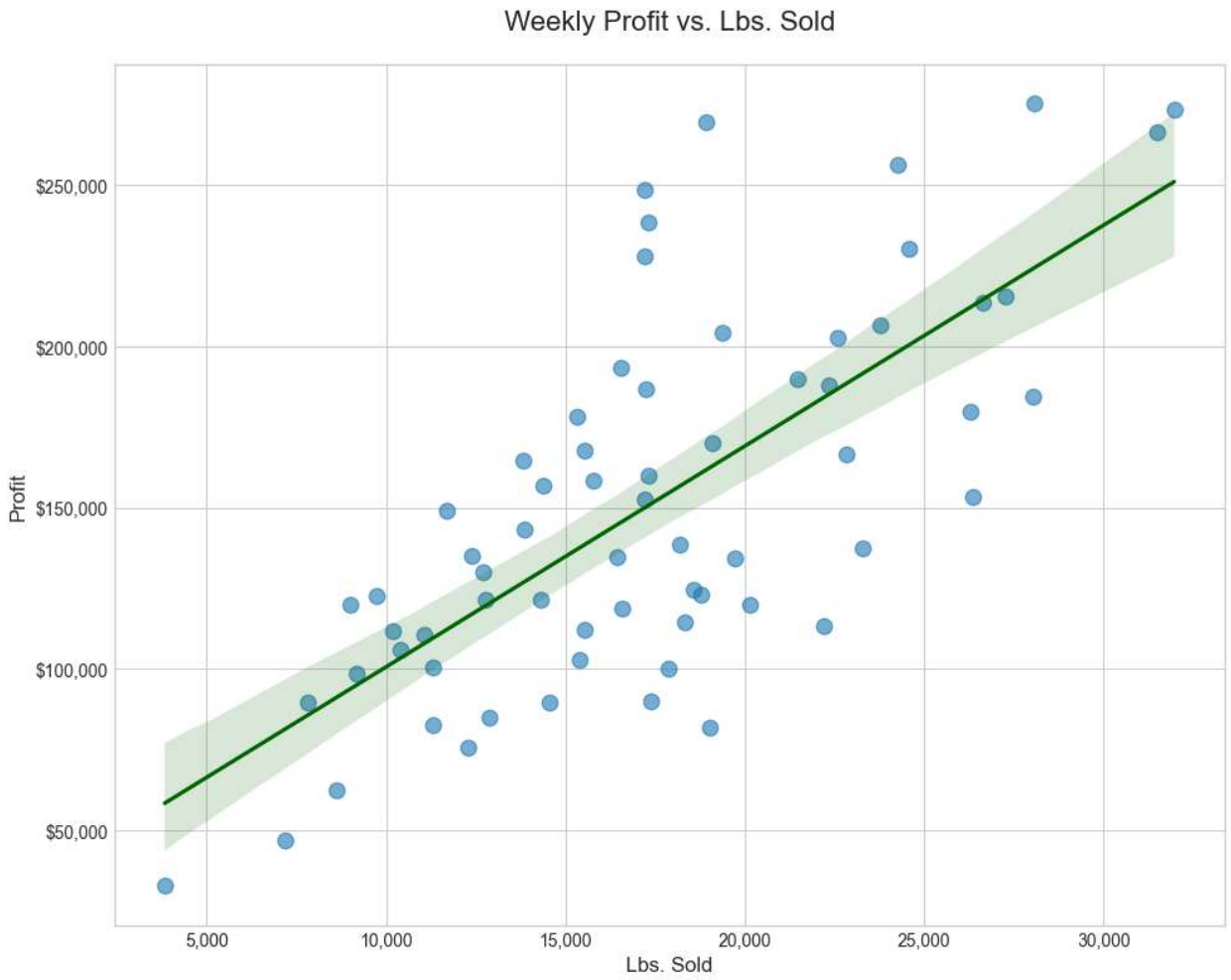


Figure 4. *Weekly Profit vs. Lbs. Sold with a regression line for estimate of Profit*

Figure 5 below displays a visual representation of lbs. sold in each period of QA's new website.

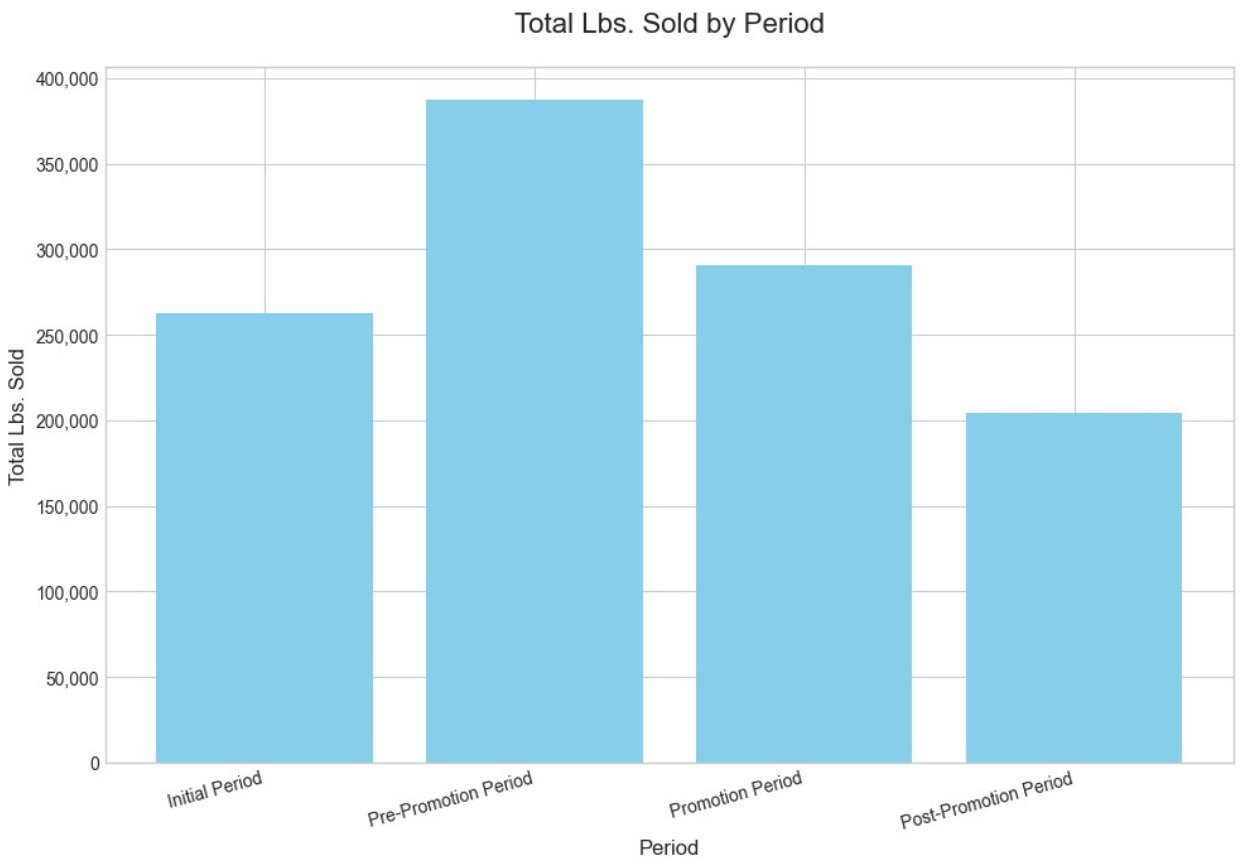


Figure 5. *Lbs. sold in each Period*

3.5 Demographics

The **Figure 7** below is a tree diagram of which countries contribute to total website views, where larger squares mean that more viewers are in that country.

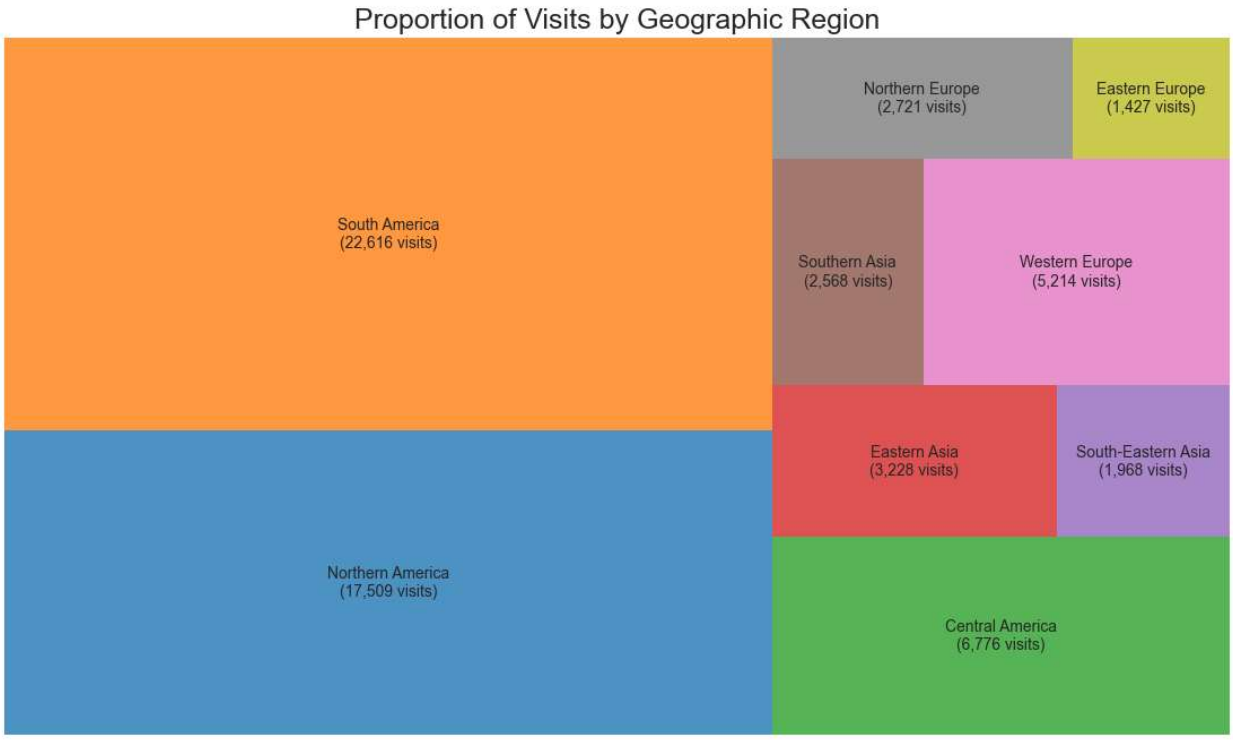


Figure 6. *Proportion of Visits by Geographic Region*

4. Conclusions

4.1 Website Visit Analysis

Observing **Figure 1**, there is a burst in both weekly and unique visits during the promotion periods, but they significantly dwindle after the promotion period ends. So, while the website garnered lots of traffic, QA was not able to keep up that momentum. Another thing **Figure 1** represents is that both the time series for weekly visits and unique visits follow each other very closely over the whole timer period for the data given. So, the fact that QA can reach that many people is very good for them, and it shows that their marketing is working and reaching different demographics. Regarding demographics, **Figure 7** displays a visual of the proportion of total visits by geographic region. Even though QA is based in the U.S., the largest portion of site visits came from South America with 22,616 visits. The North American region was not far behind with 17,509 visits.

Even though QA can reach a large audience, **Figure 2** displays the bounce rate along with new visits by period. Bounce rate denotes the percentage of people that navigate away from a site after only seeing one page and the new visits represent the percentage of people who are viewing the site for the first time. For the most part, the new visits percentage stays the same throughout

all periods but has a slight jump during the promotion period. The fact that the new visit percentage stays consistent implies that QA can draw potential customers in, but the bounce rate also surges during promotion, meaning QA cannot engage those viewers. So, it seems that the marketing and promotion is working, but once on the website there is not much that keeps viewers using it.

Metrics can also be used to analyze the success and possible improvements for QA's current strategies of marketing and their general web presence. **Table 1** shows certain metrics for each period given in the dataset. There were most page views for the site during the promotion period with a number of 52,731 which is much higher than any other period. However, even though the promotion period had the most engagement on the site, it had the least number of average pages visited per visit with just 1.79. For reference, the initial, pre-promotion, and post promotion periods had 2.28, 2.67, and 2.18 number of average pages per visit respectively. **Table 1** also displays the average amount of time a viewer spent on the site in each period, and the promotion period was the lowest with an average of just 48.8 seconds while during the pre-promotion period people would stay on the site for an average of 95.9 seconds. The bubble chart in **Figure 3** visually represents **Table 1** and it can be clearly seen that even though the promotion period had the highest number of views, it had very little pages visited per person, and people did not spend much time on the site. In **Figure 3**, the x-axis represents total page views, y-axis is the average number of pages people viewed on a visit, and the size of each bubble represents the time (s) viewers spent on the site. The different periods are also shaded according to the consistent color scheme for easy viewing. So, this validates what was found from the analysis on **Figure 2**, that while QA has great reach, their engagement falls short.

4.2 Financial Analysis

With QA being an alloy distributor, they do sell their products by the pound. **Figure 4** shows the correlation between profit and the number of pounds of alloys sold. There is a positive correlation, so the more alloys QA sells, their profit does increase. The green line on the plot denotes an estimated regression line with the green shaded areas representing the confidence interval around it. The number of pounds sold, and profit has a positive relationship, implying that if QA improves their site, it could lead to more engagement, resulting in more profit for the company.

Figure 5 has some surprising results, because QA sold less pounds after their promotion period than initially. The prepromotion period had the greatest number of pounds of alloys sold, with well over 350,000 lbs., so QA should investigate why the number of pounds sold went down significantly from the pre-promotion period to the post-promotion period, since that seems like a negative effect on their business.

4.3 Final Conclusions

Overall, the analysis conducted on the data given by QA suggests that the website had a very good reach in the beginning, but it was not sustainable, and interestingly the highest percentage of visitors were from South America. Despite this strong reach, the site showed low engagement metrics. The bounce rate surged during promotions, indicating that while people were visiting, they were quickly leaving after viewing only one page. So, QA could not keep people on the site. Even though the website did have lots of viewers, QA did not have a lasting increase in lbs. of alloys sold because the website was not engaging and was not marketed towards the right people.

5. Limitations

5.1 Data Limitations

Some limitations in the analysis include the fact that the given dataset was not very large and only covered a small amount of time. Analytics from a few years' worth of time and multiple promotional periods could have yielded more comprehensive results on how well QA's marketing strategies are working.

The data given was also only in the context of QA, it did not reflect any trends in the industrial sector or give any information on buyers and customers. The demographics data was also not connected to anything besides site visits and perhaps if it was broader to include which regions bought the most pounds or had the most inquiries then QA could create more advertisements directed towards buyers in those regions.

In the dataset, there is also no differentiation between which kinds of customers are buying alloys or making inquiries. So, it is difficult to tell if QA needs to market more towards first time or returning customers.

5.2 Limitations in the Analysis

While the analysis conducted on the data covers several possible factors for QA's success, correlation does not mean causation. So, it cannot be proven with the data given that promotion caused a drop in the number of pounds sold because there could be other factors affecting it. No conclusions made have been tested or tried by QA so there is no guarantee they would work well for this company.

6. Recommendations

Some recommendations for QA based on the results from the data analysis would be to attempt to keep visitors on the website for longer by enhancing the landing page, adding more information, and making everything more easily accessible. Also adding an online shopping cart would engage viewers more and they are more likely to peruse the site and see all options for different grades of alloys. Based on the results from the demographics analysis, QA could also

cater their marketing more towards their South American viewers, maybe by adding in Spanish articles or ads on their site. The highest number of visits also came from the referring site googleads.g.doubleclick.net, so that is a popular place from where viewers come across QA's site. So, if QA were to do the same marketing techniques they used on this site at other sites, that could also drive more traffic to the webpage.